AWS Lab: FinOps Cost Governance & Runaway Compute/Storage Triage

Duration : 90 Min.

Scenario

A company's monthly AWS bill unexpectedly surged by 45% ($18,000 overrun), triggering a budget breach alert. The FinOps engineer is given a raw Cost Explorer JSON export and resource inventory across the account. The engineer must identify the three primary cost drivers: (1) 25 unattached, orphaned gp3 and io2 EBS volumes left behind after terminated test instances, (2) five oversized p3.8xlarge GPU instances running 24/7 with 0% GPU utilization, and (3) an S3 bucket with 150 TB of un-lifecycle-managed analytics dumps. The engineer must quantify the wastage and implement remediation scripts.

Input Details

The cloud environment has been pre-configured with the following parameters: - **Platform:** AWS FinOps Cost Governance & Cost Explorer - **Cost Report:** `/opt/aws_finops/cost_explorer_report.json` ($18,000 monthly overrun) - **Cost Drivers:** (1) 25 unattached gp3/io2 EBS volumes, (2) 5 idle p3.8xlarge instances, (3) 150 TB un-lifecycle-managed S3 bucket

Username & Naming Conventions

Your candidate prefix is: **`<prefix>`** (derived from your login email before the `@` symbol, e.g. `student-demo01`). All custom resources and cloud configurations should incorporate this convention: - **Resource Naming:** Prepend or append `<prefix>` to your created AWS resources (e.g. `<prefix>-resource`). - **Candidate Tag:** Add tag `CandidatePrefix` = `<prefix>` on all managed cloud resources.

Task Objectives

Perform the following technical objectives directly in the AWS Management Console or using the AWS CLI: 1. Analyze Cost Explorer data from /opt/aws_finops/cost_explorer_report.json. 2. Identify unattached EBS volumes: query volumes where Status=available and calculate monthly idle cost. 3. Identify idle EC2 instances: correlate CPU/GPU utilization telemetry to flag oversized idle instances. 4. Identify unoptimized S3 storage: calculate cost savings from transitioning Standard to Glacier. 5. Apply and attach least-privilege IAM policy directly in AWS IAM that deletes unattached volumes older than 14 days. 6. Generate a formal FinOps executive report ~/finops_savings_plan.md detailing total identified monthly savings. 7. Implement an AWS Budget alert to notify the team when spend reaches 80% of forecasted monthly baseline.

Verification

Once completed, run the verification script to check your progress and receive your score. The verification system will inspect the live resources in the AWS cloud:

```
python run.py
```

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Data ingestion: correctly parse Cost Explorer breakdown by service and usage type	3 Marks
TC2	Orphaned storage isolation: identify all unattached EBS volumes and compute wasted monthly spend	3 Marks
TC3	Idle compute detection: identify oversized p3 instances running idle via CloudWatch metric correlation	3 Marks
TC4	S3 cost modeling: calculate exact financial savings achieved by applying Lifecycle transition rules	3 Marks
TC5	Remediation script: finops_remediation.py safely targets orphaned resources with dry-run support	3 Marks
TC6	Executive report: finops_savings_plan.md outlines immediate vs ongoing recurring savings	3 Marks
TC7	Budget guardrail: configure AWS Budget alarm threshold and SNS notification payload	2 Marks

Total Score: 20 Marks

Important Notes

- Perform all actions directly in the AWS Management Console or using the AWS CLI.
- Ensure all resource configurations, tags, and naming conventions match the requirements.
- Do not terminate or alter pre-existing cloud infrastructure outside the scope of this assessment.